

AI-Driven Sport Analytics Platform - Presentation Scripts

DIKSHANT PATEL - Script

Slides: 1-10 (Introduction, Motivation, Aim, Applications, Literature Survey Papers 1-3)

SLIDE 1: Title Slide

Good morning/afternoon everyone. I'm Dikshant Patel. Today, along with my teammates Dishant Rana, Sahil Patel, and Fenil Vachhani, we'll be presenting our project on the AI-Driven Sport Analytics Platform. This project has been developed under the guidance of Professor Karnavi Desai from the IT Department.

SLIDE 2: Outline

Before we begin, let me quickly walk you through the structure of our presentation. We'll start with an introduction to our platform, discuss our motivation and aim, explore the applications, review relevant literature, identify research gaps, outline our objectives, explain our methodology, compare existing and proposed workflows, discuss implementation details, and finally conclude with our findings and references.

SLIDE 3: Introduction

Sports analytics has fundamentally transformed how athletes train, compete, and recover. Our AI-Driven Sports Analytics Platform leverages the power of artificial intelligence and machine learning to provide comprehensive insights that help optimize performance and prevent injuries.

The key highlights of our platform include: real-time performance tracking and analysis that gives immediate feedback to athletes and coaches; predictive analytics for injury risk assessment, which helps identify potential problems before they occur; sport-specific metrics and insights tailored to different disciplines; and role-based access that ensures different stakeholders—whether players, managers, or analysts—get the information most relevant to them.

SLIDE 4: Motivation

Our motivation for developing this platform stems from four critical needs in modern sports:

First, automated insights. AI has the capability to turn complex data into quick and accurate decisions, eliminating the guesswork from coaching and training strategies.

Second, injury prevention. Through early risk detection, we can minimize injuries and reduce recovery time, keeping athletes healthier and on the field longer.

Third, personalized performance. Every athlete is different, and customized training approaches improve results while extending athletic longevity.

And fourth, competitive advantage. In today's sports landscape, data-driven strategies provide teams with a measurable edge over their competition.

SLIDE 5: Aim

The primary aim of this project is to design and develop an AI-powered sports analytics platform that enhances athletic performance and minimizes injury risks through systematic data analysis.

We focus on four core areas:

Performance Optimization—we analyze athlete performance metrics and provide personalized training recommendations for improved results.

Injury Prevention—we identify potential injury risks at an early stage using predictive analysis and support effective workload and recovery management.

Data-Driven Insights—we generate meaningful insights from sports data using analytical models and visual dashboards for better decision-making.

And User-Centric Design—we've developed a secure, role-based, and user-friendly system that's accessible across both web and mobile platforms.

SLIDE 6: Applications

Our platform serves three primary user groups:

For Players, the platform enables athletes to track personal performance, monitor health metrics, follow personalized training plans, and receive injury-risk alerts in real-time.

For Managers, it supports team oversight through performance monitoring, workload management, player comparison, and data-driven decision making that can influence game strategy and team composition.

For Data Analysts, it provides powerful tools for analyzing performance data, identifying trends, generating comprehensive reports, and supporting strategic insights that can give teams a competitive edge.

SLIDE 7: Literature Survey

We conducted an extensive literature survey to understand the current state of AI in sports analytics. We reviewed 21 research papers covering various aspects of sports analytics, machine learning applications, injury prevention, and performance optimization. Let me walk you through some of the key papers that informed our approach.

SLIDE 8: Paper 1

The first paper, "AI and Machine Learning Applications in Sports Performance Analysis," provided us with foundational knowledge about how AI and ML techniques are currently being applied in sports. This paper highlighted various algorithms and their effectiveness in analyzing athlete performance data.

SLIDE 9: Paper 2

The second paper, "Wearable Sensor Data Analytics for Athlete Performance and Injury Prevention," was particularly valuable in understanding how wearable technology can be integrated into our platform. It demonstrated how continuous monitoring through sensors can provide real-time data that's critical for both performance enhancement and injury prevention.

SLIDE 10: Paper 3

The third paper, "Predictive Analytics in Sports: Injury Risk Assessment and Performance Optimization," directly aligned with our objectives. This research provided insights into predictive modeling techniques that we could adapt for our injury risk assessment module and performance optimization algorithms.

Now, I'll hand over to Dishant, who will continue with the literature survey and discuss the research gaps we identified.

DISHANT RANA - Script

Slides: 11-22 (Literature Survey Papers 4-15, Research Gap)

SLIDE 11: Paper 4

Thank you, Dikshant. I'm Dishant Rana, and I'll continue with our literature survey.

Paper 4, "Biomechanics of the Bowling Delivery in Cricket: A Critical Analysis," provided sport-specific insights, particularly for cricket. Understanding the biomechanics helps us develop more accurate performance metrics for bowling actions, which is crucial for technique analysis and injury prevention in cricket.

SLIDE 12: Paper 5

Paper 5, "Data Science in Sports Analytics: A Review of Performance Optimization and Fan Engagement," expanded our perspective beyond just athlete performance. It showed us how data science can be applied holistically across sports, including fan engagement aspects, though our focus remains on athlete-centric analytics.

SLIDE 13: Paper 6

Paper 6, "Methodology and Evaluation in Sports Analytics: Challenges, Approaches, and Lessons Learned," was instrumental in helping us understand the methodological challenges in this domain. It highlighted common pitfalls and best practices that we incorporated into our own methodology.

SLIDE 14: Paper 7

Paper 7, "An AI-Driven Approach to Real-Time Sports Analytics and Performance Evaluation," specifically addressed real-time analytics, which is a critical component of our platform. This research helped us understand the infrastructure requirements and processing capabilities needed for real-time analysis.

SLIDE 15: Paper 8

Paper 8, "The Role of Artificial Intelligence in Sports Analytics: A Systematic Review and Meta-Analysis of Performance," provided a comprehensive overview of AI's role across multiple sports. This meta-analysis helped us identify the most effective AI techniques and their applications.

SLIDE 16: Paper 9

Paper 9, "Artificial Intelligence-Powered Sports Analytics: Enhancing Performance Through Data Science," reinforced our approach of combining AI with comprehensive data science methodologies to achieve optimal results.

SLIDE 17: Paper 10

Paper 10, "Orientation and Decision-Making for Soccer Based on Sports Analytics and AI: A Systematic Review," focused on soccer-specific applications. While our platform is designed for multiple sports, this research provided valuable insights into tactical analysis and decision-making support.

SLIDE 18: Paper 11

Paper 11, "AI-Driven Decision Support Systems for Injury Prediction and Performance Monitoring in Sports," was particularly relevant to our injury prevention objectives. It provided frameworks for building decision support systems that we adapted for our platform.

SLIDE 19: Paper 12

Paper 12, "Artificial Intelligence Applications in Sports Biomechanics: A Scoping Review of Wearables and Motion Analysis," helped us understand how to integrate wearable device data with biomechanical analysis, which is essential for comprehensive performance evaluation.

SLIDE 20: Paper 13

Paper 13, "The Role of Artificial Intelligence in Sports Performance Analysis, Injury Prevention, and Talent Identification: A Systematic Review," covered all three key areas we're addressing—performance analysis, injury prevention, and talent identification—providing a holistic view of AI applications in sports.

SLIDE 21: Paper 14

Paper 14, "Real-Time Match-Fixing Detection in Football Using Multimodal Machine Learning Models," introduced us to advanced multimodal ML approaches that we considered for our ensemble AI methodology.

SLIDE 22: Paper 15

Paper 15, "Real-Time Possessing Relationship Detection for Sports," provided insights into real-time relationship detection algorithms that could be adapted for team sports analytics within our platform.

Now let me hand over to Sahil, who will continue with the remaining papers and discuss our research gap, objectives, and methodology.

SAHIL PATEL - Script

Slides: 23-34 (Literature Survey Papers 16-21, Research Gap, Objectives, Methodology, Workflows)

SLIDE 23: Paper 16

Thank you, Dishant. I'm Sahil Patel, and I'll continue from here.

Paper 16, "Scalable Infrastructure for Efficient Real-Time Sports Analytics," addressed the infrastructure challenges we would face when implementing real-time analytics at scale. This research influenced our backend architecture decisions.

SLIDE 24: Paper 17

Paper 17, "Analysis of Cricket Ball Type and Innings on State Level Cricket Batter's Performance," provided cricket-specific insights that helped us understand how environmental and equipment factors affect performance, which we incorporated into our multi-dimensional analysis approach.

SLIDE 25: Paper 18

Paper 18, "Data Analytics in the Game of Cricket: A Novel Paradigm," introduced innovative approaches to cricket analytics that we adapted for our sport-specific metrics module.

SLIDE 26: Paper 19

Paper 19, "Applications of Machine Learning in Cricket: A Systematic Review," gave us a comprehensive understanding of ML applications specific to cricket, helping us develop more accurate cricket-focused features.

SLIDE 27: Paper 20

Paper 20, "Deep Learning in Cricket: A Comprehensive Survey of Shot Detection and Performance Analysis," explored deep learning techniques for cricket shot detection and performance analysis, which we incorporated into our video analysis capabilities.

SLIDE 29: Paper 21

Finally, Paper 21, "Artificial intelligence in sport: A narrative review of applications, challenges and future trends," provided us with insights into future trends and challenges in AI-driven sports analytics, helping us design a platform that's future-ready and addresses current limitations.

SLIDE 30: Research Gap

After this extensive literature review, we identified three critical research gaps:

First, there's a lack of real-time AI-driven analytics that integrate both performance analysis and injury prediction in a single platform. Most existing solutions focus on one or the other, but not both simultaneously.

Second, we found limited role-based platforms that successfully combine insights for athletes, managers, and analysts in one unified system. Most platforms serve only one user type.

Third, there's insufficient use of predictive machine learning models for proactive injury prevention. Most systems are reactive rather than predictive.

SLIDE 31: Objectives

Based on these gaps, we established three primary objectives:

One, to develop an AI-powered sports analytics platform specifically designed for performance optimization.

Two, to implement injury risk prediction using machine learning and historical data analysis.

And three, to provide role-based dashboards with actionable data insights tailored to each user type—players, managers, and data analysts.

SLIDE 32: Methodology

Our methodology is built on four key pillars:

Holistic Data Integration—we combine physical, tactical, psychological, environmental, and historical data to ensure complete and accurate performance analysis. This comprehensive approach gives us a fuller picture than single-source data systems.

Multi-Dimensional Analysis—performance and injury risk are evaluated using multiple weighted factors instead of relying on a single metric. This provides more nuanced and accurate assessments.

Ensemble and Adaptive AI—we use multiple AI models working together to improve prediction accuracy beyond what traditional single-model systems can achieve. This ensemble approach reduces error and increases reliability.

Continuous Learning and Explainability—our system learns from real outcomes over time and clearly explains its predictions to ensure trust and enable better decision-making by users.

SLIDE 33: Existing Workflow

Let's look at the existing workflow in traditional sports analytics. As you can see from this diagram, the current approach typically involves manual data collection, separate analysis tools, limited integration between different data sources, and delayed insights that may come too late to prevent injuries or optimize training.

SLIDE 34: Proposed Workflow

In contrast, our proposed workflow shows a fully integrated system where data from multiple sources—wearables, performance metrics, video analysis, and health records—flows seamlessly into our AI engine. This engine processes data in real-time, applies our ensemble machine learning models for both performance optimization and injury prediction, and delivers instant, actionable insights to all stakeholders through their role-specific dashboards.

Now, I'll hand over to Fenil, who will discuss the implementation details and our conclusion.

FENIL VACHHANI - Script

Slides: 35-41 (Implementation, Conclusion, References, Thank You)

SLIDE 35: Implementation

Thank you, Sahil. I'm Fenil Vachhani, and I'll walk you through our implementation approach.

For the Frontend, we've used React.js for building a responsive and dynamic user interface, Tailwind CSS for modern, clean styling, and Chart.js for creating interactive dashboards and data visualizations that make complex data easy to understand.

For the Backend, we've implemented Node.js and Express.js to create a robust server architecture, built RESTful APIs for seamless communication between frontend and backend, and integrated JWT authentication

to ensure secure, role-based access control.

For the AI Module, we've developed Python-based machine learning models that handle both prediction tasks—injury risk assessment and performance recommendations. These models are trained on historical data and continuously updated as new data comes in, ensuring they remain accurate and relevant.

SLIDE 36: Conclusion

In conclusion, our AI-Driven Sports Analytics Platform represents a significant advancement in sports technology.

The platform enables truly data-driven decision-making in sports by providing comprehensive, real-time insights that were previously unavailable or required manual analysis.

It directly improves athlete performance through personalized training recommendations and helps prevent injuries through predictive risk assessment, keeping athletes healthier and performing at their best.

Most importantly, it's a scalable solution that can be adapted for multiple sports and serves various stakeholders—from individual athletes to large sports organizations. The role-based architecture ensures everyone gets the information they need in the format they need it.

SLIDE 37: References - Part 1

Our research was built on a solid foundation of academic literature. Here are the first set of references that informed our work:

Starting with Mandot and Chawla's 2013 work on artificial intelligence-based integrated cricket coaching, Hassan et al.'s 2024 work on PlayerRank using learning-to-rank AI for cricket player positioning, Agrawal and Kandhway's 2024 research on winner prediction in ongoing cricket matches, Thander's 2023 analysis of bowling biomechanics in cricket, and Obi et al.'s 2024 review of data science in sports analytics covering performance optimization and fan engagement.

SLIDE 38: References - Part 2

Continuing with our references: Davis et al.'s comprehensive 2024 work on methodology and evaluation in sports analytics, Mohana et al.'s 2024 research on AI-driven approaches to real-time sports analytics, Pietraszewski et al.'s systematic review and meta-analysis of AI's role in sports analytics, Wbaid et al.'s 2025 work on AI-powered sports analytics, and Pu et al.'s 2024 systematic review on orientation and decision-making in soccer based on sports analytics and AI.

SLIDE 39: References - Part 3

Additional references include: Jain's 2025 work on deep learning models for player performance analysis and injury prediction, Souaifi et al.'s scoping review on AI in sports biomechanics with a focus on wearable technology, Pisaniello's 2024 analysis of how AI is transforming sports performance and strategy, Kim et al.'s

work on AI-based betting anomaly detection, and Xu and Peng's 2020 research on real-time possessing relationship detection for sports analytics.

SLIDE 40: References - Part 4

Our final set of references includes: Johansen et al.'s 2020 work on scalable infrastructure for real-time sports analytics, Connor et al.'s analysis of cricket ball type and innings effects on performance, Vishwarupe et al.'s novel paradigm for data analytics in cricket, Wickramasinghe's systematic review of machine learning applications in cricket, Mahajan et al.'s comprehensive survey of deep learning in cricket, and Zhou et al.'s 2025 narrative review of AI in sport covering applications, challenges, and future trends.

SLIDE 41: Thank You

Thank you all for your attention. We're now open to any questions you may have about our AI-Driven Sports Analytics Platform.

Q&A Preparation Tips for All Members:

Dikshant - Be prepared to answer about:

- Overall platform architecture
- Motivation and real-world applications
- User roles and access control

Dishant - Be prepared to answer about:

- Literature review methodology
- How specific papers influenced design choices
- Research gap identification process

Sahil - Be prepared to answer about:

- Methodology details
- Machine learning models used
- Workflow differences between existing and proposed systems
- Multi-dimensional analysis approach

Fenil - Be prepared to answer about:

- Technical implementation specifics
- Technology stack choices

- Scalability considerations
- Future enhancements
- API design and integration

All Members - Be ready to discuss:

- Data privacy and security
- Model accuracy and validation
- Integration with existing sports infrastructure
- Cost-benefit analysis
- Timeline for deployment
- Specific sport adaptations